Choices, limits and a goal

Optimization · Level 2 · Lesson 1



In this lesson you will

- Write choices as variables with units.
- Write limits as inequalities and the goal as an expression.
- Check whether a plan fits by putting its numbers in.

Key ideas

- Variables: letters for choices
- At least ≥, at most ≤
- Goal: make it small or large
- Feasible: keeps every limit

Activity: The trip snack model

1. Choose (3 min): Pick three snacks and their prices.
2. Write (7 min): Variables, two limits and a goal.
3. Check (7 min): Try three plans; feasible or not?
4. Share (3 min): Swap with a partner: what rule is missing?

Use small, round prices so you can check plans in your head. No shop nearby? Make up the prices.

Let’s discuss

1. Why are prices data and not variables?
2. When is the cheapest plan not the best one?
3. Which rule did your partner’s model forget?

Exit ticket

- One limit from my model, written as an inequality.